

TEST PLAN

VWO Login Dashboard

app.vwo.com

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1. Introduction

1.1 Purpose

This Test Plan defines the testing strategy, scope, objectives, resources, schedule, and deliverables for validating the VWO (Visual Website Optimizer) Login Dashboard at app.vwo.com. The plan ensures the login system meets all functional, security, performance, and usability requirements defined in the Product Requirements Document (PRD).

1.2 Project Background

VWO is a leading digital experience optimization platform used by 4,000+ brands across 90 countries. The login dashboard is the critical entry point for all users. Any defect in this component directly impacts user access, security, and business continuity.

1.3 Document Scope

This document covers end-to-end testing of the VWO Login Dashboard including authentication flows, password management, SSO, UI/UX, accessibility, performance, and security aspects.

2. Test Objectives

- Verify all authentication flows (email/password, SSO, 2FA) work correctly.
- Validate all input field validations and error message accuracy.
- Ensure password management (forgot password, reset) functions end-to-end.
- Confirm UI/UX compliance with design specs (Light/Dark mode, responsive design).
- Validate accessibility compliance to WCAG 2.1 AA standards.
- Verify security controls: encryption, rate limiting, session management, brute force protection.
- Confirm performance targets: page load < 2 seconds, 99.9% uptime.
- Validate integration with VWO core platform and third-party identity providers.
- Ensure GDPR/CCPA compliance for user data handling during authentication.

3. Test Scope

3.1 In Scope

- Login form: email/password authentication
- Input validation: real-time and on-blur field validation
- Remember Me functionality
- Forgot Password / Password Reset flow
- Multi-Factor Authentication (2FA)
- Single Sign-On (SSO) - SAML and OAuth
- Social login (Google, Microsoft)
- Session management and timeout
- Error handling and user messaging
- Responsive design (desktop, tablet, mobile)
- Light Mode and Dark Mode theme switching
- Accessibility features (keyboard navigation, ARIA labels, screen reader)
- Security: HTTPS enforcement, rate limiting, brute force protection

- Performance: page load time and concurrent user handling
- Post-login redirect to VWO dashboard
- New user registration / free trial signup path

3.2 Out of Scope

- VWO core platform features beyond the login dashboard
- Backend infrastructure provisioning and DevOps pipelines
- Third-party identity provider internal systems (Google, Microsoft)
- Biometric authentication (future enhancement)
- Adaptive / risk-based authentication (future enhancement)
- Progressive Web App (PWA) features (future enhancement)

4. Test Approach & Strategy

Test Type	Description
Functional Testing	Validate every feature against PRD acceptance criteria. Cover positive, negative, and boundary cases.
Integration Testing	Verify login flow integration with VWO core platform, SSO providers, and social login APIs.
Security Testing	Input injection, session hijacking attempts, brute force simulation, and SSL/TLS validation.
Performance Testing	Load test with concurrent users; measure login page load time and response under stress.
Usability Testing	Validate UI against design specs: Light/Dark themes, responsive layouts, and user flow.
Accessibility Testing	Automated (Axe, WAVE) + manual keyboard/screen reader testing against WCAG 2.1 AA.
Regression Testing	Re-run full suite on every build to ensure no existing functionality is broken.
Exploratory Testing	Ad-hoc sessions to discover defects outside scripted scenarios.

5. Test Cases

Priority: P1 = Critical, P2 = High, P3 = Medium. Status column to be updated during test execution.

5.1 Authentication - Login Flow

TC ID	Test Case Title	Steps	Expected Result	Priority	Status
TC-AUTH-01	Valid Login	1. Open app.vwo.com 2. Enter valid email & password 3. Click Login	Redirected to VWO dashboard. Session created.	P1	Not Run
TC-AUTH-02	Invalid Password	1. Enter valid email 2. Enter wrong password 3. Click Login	Error: "Incorrect email or password." No redirect.	P1	Not Run
TC-AUTH-03	Invalid Email Format	1. Enter "notanemail" 2. Click Login	Inline error: "Please enter a valid email address."	P1	Not Run

TC-AUTH-04	Empty Email Field	1. Leave email blank 2. Enter password 3. Click Login	Validation error on email field. Login blocked.	P1	Not Run
TC-AUTH-05	Empty Password Field	1. Enter valid email 2. Leave password blank 3. Click Login	Validation error on password field. Login blocked.	P1	Not Run
TC-AUTH-06	Both Fields Empty	1. Click Login with no credentials entered	Validation errors on both fields shown simultaneously.	P1	Not Run
TC-AUTH-07	Remember Me - Checked	1. Login with valid creds 2. Check Remember Me 3. Close & reopen browser	User remains logged in or credentials pre-filled.	P2	Not Run
TC-AUTH-08	Remember Me - Unchecked	1. Login without Remember Me 2. Close & reopen browser	User is logged out. No credentials pre-filled.	P2	Not Run
TC-AUTH-09	Auto-focus on Email Field	1. Open login page	Cursor automatically focused on email input.	P3	Not Run
TC-AUTH-10	Clickable Form Labels	1. Click on the "Email" label text	Focus moves to email input field.	P3	Not Run

5.2 Password Management

TC ID	Test Case Title	Steps	Expected Result	Priority	Status
TC-PWD-01	Forgot Password Link Present	1. View login page	"Forgot Password" link is visible and clickable.	P1	Not Run
TC-PWD-02	Forgot Password - Valid Email	1. Click Forgot Password 2. Enter registered email 3. Submit	Success message shown. Reset email sent to inbox.	P1	Not Run
TC-PWD-03	Forgot Password - Unregistered Email	1. Enter unregistered email 2. Submit	Generic message: "If this email exists, a reset link has been sent."	P1	Not Run
TC-PWD-04	Reset Link - Valid Token	1. Click reset link from email within expiry time	Reset password form opens. Token accepted.	P1	Not Run
TC-PWD-05	Reset Link - Expired Token	1. Use an expired password reset link	Error: "This link has expired. Please request a new one."	P1	Not Run
TC-PWD-06	New Password - Meets Requirements	1. Enter a compliant password 2. Confirm it 3. Submit	Password updated. Redirected to login with success message.	P1	Not Run
TC-PWD-07	New Password - Too Weak	1. Enter weak password (e.g. "123") 2. Submit	Inline error showing which password requirements are not met.	P2	Not Run
TC-PWD-08	Password Strength Indicator	1. Start typing a new password	Strength indicator updates in real time (Weak/Medium/Strong).	P2	Not Run
TC-PWD-	Reset Token	1. Use a valid reset link	Second use rejected:	P1	Not Run

09	Single Use	2. Try using the same link again	"Link already used or expired."		
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5.3 Multi-Factor Authentication (2FA)

TC ID	Test Case Title	Steps	Expected Result	Priority	Status
TC-MFA-01	2FA - Valid OTP	1. Login with 2FA-enabled account 2. Enter correct OTP	User authenticated and redirected to dashboard.	P1	Not Run
TC-MFA-02	2FA - Invalid OTP	1. Login with 2FA account 2. Enter wrong OTP	Error: "Invalid code. Please try again."	P1	Not Run
TC-MFA-03	2FA - Expired OTP	1. Wait for OTP to expire 2. Enter expired OTP	Error: "Code has expired. Please request a new one."	P1	Not Run
TC-MFA-04	2FA - Resend Code	1. Click "Resend Code" on 2FA screen	New OTP sent. Confirmation message shown. Previous OTP invalidated.	P2	Not Run
TC-MFA-05	2FA - Bypass Attempt	1. Login 2. Skip 2FA step 3. Access dashboard URL directly	Redirected back to 2FA verification. Dashboard access denied.	P1	Not Run

5.4 Single Sign-On (SSO)

TC ID	Test Case Title	Steps	Expected Result	Priority	Status
TC-SSO-01	SAML SSO - Valid Login	1. Click SSO login 2. Enter org domain 3. Complete IdP authentication	Redirected to VWO dashboard. Session established.	P1	Not Run
TC-SSO-02	OAuth SSO - Valid Login	1. Initiate OAuth SSO 2. Authenticate with OAuth IdP	Tokens exchanged successfully. User logged in.	P1	Not Run
TC-SSO-03	SSO - Invalid Domain	1. Enter an unregistered SSO domain	Error: "SSO is not configured for this domain."	P1	Not Run
TC-SSO-04	Social Login - Google	1. Click "Continue with Google" 2. Select a Google account	User authenticated via Google and redirected to dashboard.	P2	Not Run
TC-SSO-05	Social Login - Microsoft	1. Click "Continue with Microsoft" 2. Select Microsoft account	User authenticated and redirected to dashboard.	P2	Not Run
TC-SSO-06	SSO Session Expiry	1. Login via SSO 2. Allow session to expire	User redirected to login with session expired message.	P2	Not Run

5.5 Session Management

TC ID	Test Case Title	Steps	Expected Result	Priority	Status
TC-SES-	Session	1. Login	Auto logout.	P1	Not Run

01	Timeout on Inactivity	2. Leave idle beyond timeout threshold	Redirected to login with timeout message.		
TC-SES-02	Session Persists During Active Use	1. Login 2. Actively use the platform continuously	Session remains active. No unexpected logout.	P1	Not Run
TC-SES-03	Logout Functionality	1. Click Logout	Session destroyed. Redirected to login. Back button cannot restore session.	P1	Not Run
TC-SES-04	Concurrent Sessions Same Account	1. Login on Browser A 2. Login on Browser B same account	Both active OR earlier session invalidated per policy.	P2	Not Run
TC-SES-05	Session Cookie Security Flags	1. Inspect session cookie in browser DevTools	Cookie has Secure, HttpOnly, and SameSite=Strict flags set.	P1	Not Run

5.6 Security Testing

TC ID	Test Case Title	Steps	Expected Result	Priority	Status
TC-SEC-01	HTTPS Enforcement	1. Navigate to http://app.vwo.com (HTTP)	Automatically redirected to HTTPS. No data sent over HTTP.	P1	Not Run
TC-SEC-02	SQL Injection - Email Field	1. Enter: ' OR '1'='1 in email 2. Submit	Input sanitised. No SQL error. Standard error message shown.	P1	Not Run
TC-SEC-03	XSS - Input Fields	1. Enter: <script>alert(1)</script> in email 2. Submit	Input escaped. No script executes. Standard error shown.	P1	Not Run
TC-SEC-04	Brute Force Protection	1. Attempt 10+ failed logins rapidly	Account locked / CAPTCHA triggered after threshold. Lockout message shown.	P1	Not Run
TC-SEC-05	Rate Limiting	1. Send rapid repeated login requests via HTTP client	Requests throttled after threshold. HTTP 429 returned.	P1	Not Run
TC-SEC-06	Password Not Exposed in Transit	1. Capture network traffic during login using browser DevTools	Password not visible in plain text in request payload or URL.	P1	Not Run
TC-SEC-07	CSRF Protection	1. Attempt cross-site form submission without valid CSRF token	Request rejected. CSRF validation blocks forged request.	P1	Not Run
TC-SEC-08	SSL Certificate Valid	1. Check SSL certificate on app.vwo.com	Valid certificate. No browser warnings. TLS 1.2+ enforced.	P1	Not Run

5.7 UI / UX & Responsive Design

TC ID	Test Case Title	Steps	Expected Result	Priority	Status
TC-UI-01	Desktop Layout (1920x1080)	1. Open login page on desktop browser	All elements visible, aligned, and styled correctly.	P1	Not Run
TC-UI-02	Tablet Layout (768x1024)	1. Open login page on tablet viewport	Responsive layout. No overflow. Touch targets adequate.	P2	Not Run
TC-UI-03	Mobile Layout (375x812)	1. Open on mobile browser	Mobile-optimized layout. Email keyboard type auto-triggered.	P2	Not Run
TC-UI-04	Dark Mode	1. Switch to Dark Mode 2. View login page	All elements render correctly in dark theme. Readable text.	P2	Not Run
TC-UI-05	Light Mode	1. Switch to Light Mode 2. View login page	All elements render correctly in light theme.	P2	Not Run
TC-UI-06	Loading State on Submit	1. Submit login form 2. Observe during server processing	Spinner/loading state shown. Form disabled to prevent duplicate submissions.	P2	Not Run
TC-UI-07	Free Trial / Sign Up CTA	1. View login page for Sign Up / Free Trial link	Link is clearly visible and navigates to registration page.	P2	Not Run
TC-UI-08	VWO Branding Consistency	1. Review login page visuals	Colors, logo, and typography match VWO brand guidelines.	P3	Not Run

5.8 Accessibility (WCAG 2.1 AA)

TC ID	Test Case Title	Steps	Expected Result	Priority	Status
TC-ACC-01	Keyboard Navigation	1. Tab through all form elements 2. Submit using Enter key only	All elements reachable via Tab. Logical focus order. Login works without mouse.	P1	Not Run
TC-ACC-02	Screen Reader Compatibility	1. Enable NVDA or VoiceOver 2. Navigate login page	All labels, errors, and buttons announced correctly.	P1	Not Run
TC-ACC-03	ARIA Labels Present	1. Inspect DOM for ARIA attributes	All inputs have aria-label or aria-labelledby. Errors have aria-live regions.	P1	Not Run
TC-ACC-04	Colour Contrast Ratio	1. Run contrast checker on all text/background combinations	Minimum 4.5:1 for normal text; 3:1 for large text. WCAG AA met.	P2	Not Run
TC-ACC-05	High Contrast Mode	1. Enable OS High Contrast Mode 2. View login page	Page remains fully usable. No elements disappear or become	P2	Not Run

			unreadable.		
TC-ACC-06	Visible Focus Indicators	1. Tab through all interactive elements	Clearly visible focus ring on every focusable element.	P2	Not Run

5.9 Performance Testing

TC ID	Test Case Title	Steps	Expected Result	Priority	Status
TC-PERF-01	Page Load - Standard Connection	1. Open login page on broadband 2. Measure full page load time	Page fully loaded within 2 seconds.	P1	Not Run
TC-PERF-02	Page Load - 3G Network	1. Throttle network to 3G 2. Open login page	Page loads within 5 seconds. No broken assets.	P2	Not Run
TC-PERF-03	Concurrent Users Load Test	1. Simulate 1,000 concurrent login attempts	System handles load. Response time < 3s. No 500 errors.	P1	Not Run
TC-PERF-04	Stress Test Beyond Capacity	1. Gradually increase load beyond 1,000 concurrent users	System degrades gracefully. No data loss or security failures.	P2	Not Run
TC-PERF-05	Login API Response Time	1. Measure server response time for POST /login request	API responds within 500ms under normal load.	P2	Not Run
TC-PERF-06	Uptime Monitoring	1. Monitor login page availability over 24 hours	99.9% uptime achieved. No unplanned downtime.	P1	Not Run

5.10 Integration Testing

TC ID	Test Case Title	Steps	Expected Result	Priority	Status
TC-INT-01	Post-Login Redirect to Dashboard	1. Login with valid credentials	User immediately redirected to personalised VWO dashboard.	P1	Not Run
TC-INT-02	Login Success Event Tracked	1. Login successfully 2. Check analytics events	Login success event captured in analytics platform.	P2	Not Run
TC-INT-03	Login Failure Event Tracked	1. Fail login attempt 2. Check analytics events	Login failure event logged with error type.	P2	Not Run
TC-INT-04	Support Link Accessible from Login Page	1. Click help/support link on login page	Navigated to support portal or chat widget opens.	P3	Not Run
TC-INT-05	SSO Session Handoff to Core Platform	1. Login via SSO 2. Navigate to a VWO feature	Session token accepted by core platform. No re-authentication required.	P1	Not Run

6. Test Environment

6.1 Browsers & Platforms

Browser	Version	Platform
Chrome	Latest stable	Desktop + Android Mobile
Firefox	Latest stable	Desktop
Safari	Latest stable	Desktop + iOS Mobile
Edge	Latest stable	Desktop

6.2 Test Data Requirements

- Valid registered accounts (standard user, admin, enterprise SSO user)
- Invalid credentials (wrong password, unregistered email)
- Accounts with 2FA enabled
- Accounts with SSO configured (SAML and OAuth)
- Expired password reset tokens
- Test SAML IdP and OAuth app configuration on staging

6.3 Environments

- Staging: <https://staging.app.vwo.com> (primary test environment)
- Production: <https://app.vwo.com> (smoke tests only post-release)

7. Entry & Exit Criteria

7.1 Entry Criteria

- Login dashboard deployed and stable on staging environment
- PRD reviewed and signed off by Product Owner
- Test cases reviewed and approved by QA Lead
- Test data and test accounts provisioned
- Test environment smoke-tested and verified as stable

7.2 Exit Criteria

- 100% of P1 (Critical) test cases executed and passed
- 95%+ of P2 (High) test cases executed and passed
- Zero open P1 defects at time of sign-off
- All security test cases passed with no critical findings
- Performance benchmarks met (< 2s page load, 99.9% uptime)
- WCAG 2.1 AA accessibility compliance verified
- Test Summary Report reviewed and signed off by QA Lead and Product Owner

8. Defect Management

8.1 Severity Levels

Severity	Definition	Resolution SLA
P1 - Critical	Login completely broken; security	< 4 hours

	vulnerability; data loss risk	
P2 - High	Major feature not working; significant user experience failure	< 1 business day
P3 - Medium	Feature partially broken; workaround exists	< 3 business days
P4 - Low	Minor UI/cosmetic issue; no functional impact	Next sprint

All defects will be logged in the project issue tracker with: Test Case ID, Steps to Reproduce, Expected vs Actual Result, Screenshots/Logs, Environment details, and Severity.

9. Risks & Mitigations

Risk	Likelihood	Impact	Mitigation
SSO IdP unavailable in staging	Medium	High	Set up mock SAML/OAuth IdP for isolated testing
Test environment instability	Medium	High	Daily health checks; rollback plan in place
Incomplete test data setup	Low	High	Prepare test data scripts before test execution
Security vulnerabilities found late	Low	Critical	Run security tests in Phase 1, not just at end
Performance degradation under load	Medium	High	Early load testing; involve DevOps for scaling
Browser/device compatibility issues	Medium	Medium	Test on all target browsers before sign-off
Accessibility tool coverage gaps	Low	Medium	Combine automated tools with manual screen reader sessions

10. Test Schedule

Phase	Activities	Duration	Owner
Phase 1 - Setup	Environment setup, test data prep, test case review	2 days	QA Lead
Phase 2 - Functional	TC-AUTH, TC-PWD, TC-MFA, TC-SSO, TC-SES execution	3 days	QA Team
Phase 3 - Security	TC-SEC execution, penetration test basics	2 days	QA + Security
Phase 4 - UI/UX & A11y	TC-UI, TC-ACC execution across browsers and devices	2 days	QA Team
Phase 5 - Performance	TC-PERF load and stress testing	1 day	QA + DevOps
Phase 6 - Integration	TC-INT execution, SSO end-to-end validation	1 day	QA Team
Phase 7 - Regression	Full regression run on final build candidate	2 days	QA Team

Phase 8 - Sign-off	Defect review, test summary report, stakeholder sign-off	1 day	QA Lead + PO
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11. Test Deliverables

- Test Plan Document (this document)
- Test Cases Spreadsheet (detailed steps, test data, expected results)
- Defect Reports (logged per severity in issue tracker)
- Test Execution Report (pass/fail summary per test cycle)
- Performance Test Report (load and stress test results)
- Accessibility Audit Report (WCAG 2.1 AA compliance status)
- Security Test Report (findings and remediation status)
- Final Test Summary Report (sign-off document)

12. Success Metrics & KPIs

The following metrics define test success aligned to PRD KPIs:

Metric	Target	Category
Login Success Rate	> 95% successful authentications	Performance
Page Load Time	< 2 seconds on standard connection	Performance
P1 Test Cases Pass Rate	100%	Quality
P2 Test Cases Pass Rate	>= 95%	Quality
Security Incidents	Zero successful brute force / hijacking	Security
WCAG 2.1 AA Compliance	100% of critical UI components	Accessibility
System Uptime	99.9% during test period	Reliability
Login-Related Support Tickets	Reduced by 20% post-release	Business

13. Approvals & Sign-off

Role	Name	Signature	Date
QA Lead			
Product Owner			
Dev Lead			
Security Lead			